

Advanced Certification in

Applied Data Science, Machine Learning and IoT



Electronics & ICT Academy
Indian Institute of Technology Guwahati
An Initiative of Ministry of Electronics & Information Technology(Meity)



इलेक्ट्रॉनिक्स एवं
सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF
**ELECTRONICS AND
INFORMATION TECHNOLOGY**

Course Curriculum

Duration: 9 Months (240Hrs)

DS | ML | DL | AI | Generative AI

In Collaboration With

**The IoT
Academy**

Connecting The Unconnected

ABOUT THE IIT GUWAHATI



Indian Institute of Technology Guwahati, the sixth member of the IIT fraternity, was established in 1994. The academic programme of IIT Guwahati commenced in 1995. At present the Institute has eleven departments and three inter-disciplinary academic centres covering all the major engineering, science and humanities disciplines, offering BTech, BDes, MA, MDes, MTech, MSc and PhD programmes. Within a short period of time, IIT Guwahati has been able to build up world class infrastructure and a reputation for itself.

ABOUT

E&ICT ACADEMY, IIT GUWAHATI



Electronics & ICT Academy

Indian Institute of Technology Guwahati

An Initiative of Ministry of Electronics & Information Technology (MeitY)

Electronics and ICT Academy aims to provide specialized training to the faculties of Engineering, Arts, Commerce, Science colleges and Polytechnics institutes by developing short term training programmes on fundamental and advanced topics in IT, Electronics & Communication, Product Design, Manufacturing. In addition, the Academy conducts specialized customized training programmes and research promotion workshops for corporate sector & educational institutions.



Data Analytics, Data Science & Visualization with Python

Module-1: Python Basics

- ✓ What is Python
- ✓ Application of Python
- ✓ Why use Python for AI-ML Installation Anaconda/Other Idle
- ✓ Python Tokens
- ✓ Data Types in Python
- ✓ Conditional Statement
- ✓ Loops in Python
- ✓ Functions in Python
- ✓ Advance Functions
- ✓ File Handling
- ✓ **Mini Project**

Module-2: Intro to Git & Github and Kaggle

- ✓ Git and GitHub
- ✓ kaggle

Module-3: Python NumPy & Pandas

- ✓ Introduction to NumPy
- ✓ Exploring a NumPy Array
- ✓ Indexing & Slicing a NumPy Array
- ✓ Manipulating a NumPy Array
- ✓ Performing Mathematical & Statistical Functions using NumPy
- ✓ Performing Linear Algebra Operations using NumPy
- ✓ Introduction to Pandas
- ✓ Exploring Pandas Series
- ✓ Introduction to Pandas DataFrame
- ✓ Importing & Exporting Data
- ✓ Implementing basic DataFrame functionalities
- ✓ Exploring Descriptive Statistics with Pandas
- ✓ **Mini Project**

Modules-4: R Language Essentials

- ✓ Fundamentals of R
- ✓ Vectors & control statements
- ✓ Functions in R
- ✓ Matrices & strings
- ✓ Lists & arrays in R
- ✓ Data visualization in R

Modules-5: Exploratory Data Analysis (EDA)

- ✓ Data Types
- ✓ Dispersion & Skewness
- ✓ Data imputation
- ✓ Data Pre-processing
- ✓ Data Cleaning
- ✓ Data Manipulation
- ✓ Advanced Manipulation
- ✓ **Mini Project**

Module-6: SQL For Data Analytics

- ✓ SQL Basic
- ✓ SQL Joins
- ✓ SQL Aggregations
- ✓ Subqueries and Temp Tables
- ✓ SQL Data Cleaning
- ✓ Window Functions
- ✓ **Mini Project**

Module-7: Data Analysis with Excel

- ✓ Key Formulas and Functions, Ranges and Tables
- ✓ Data Cleaning – Text Functions, Dates and Times
- ✓ Conditional Formatting, Sorting and Filtering
- ✓ Dashboard Creation
- ✓ Analysis with Pivot Tables
- ✓ Data Analysis in Excel – Trends and Patterns
- ✓ Data Visualization in Excel – Charts and Plots
- ✓ Functions in Python
- ✓ Advance Functions
- ✓ File Handling
- ✓ **Mini Project**

Module-8: Data Visualization with Python

- ✓ Why Data Visualization?
- ✓ Introduction to Data Visualization
- ✓ Libraries & Tools for Data Visualization in Python
- ✓ Static Data Visualization Using Seaborn
- ✓ Interactive Data Visualization Using Plotly Express
- ✓ Interactive Animations & Facet Plots

Module-9: Tableau for Business Intelligence

- ✓ Understanding Data
- ✓ Creating Your First visualization
- ✓ Tableau Calculations
- ✓ Formatting Visualizations
- ✓ Manipulating Data in Tableau
- ✓ Creating Dashboards AND Stories
- ✓ Distributing & Publishing Your Visualization
- ✓ **Mini Project**

Module-10: Visual Storytelling using Power BI

- ✓ Introduction To Power BI
- ✓ Creating Power BI Reports, Auto Filters
- ✓ Report Visualization And Properties
- ✓ Chart And Map Report Properties
- ✓ Hierarchies And Drilldown Reports
- ✓ Power BI Deployment & Cloud
- ✓ Improving Power BI Reports And More...
- ✓ **Mini Project**

Applied Maths, Statistics and Probability for ML & AI

Module-1: Mathematics for Machine Learning & AI

- ✓ Linear Algebra
- ✓ Multi-variable Calculus
- ✓ Introduction to Calculus

Module-2: Statistics for Machine Learning & AI

- ✓ Applications of Statistics
- ✓ Sampling techniques
- ✓ Introduction to Statistics
- ✓ Descriptive Statistics
- ✓ Categories of Data
- ✓ Measure Used in Descriptive Statistics
- ✓ Basic Terminologies in Statistics
- ✓ Z-Scores

Module-3: Probability

- ✓ What is Probability?
- ✓ Random Variables
- ✓ Rules of probability
- ✓ Probability Distribution Functions
- ✓ Types of Probability
- ✓ Mini Project

Module-4: Inferential Statistics

- ✓ Introduction to Inferential Statistics
- ✓ One Sample Z test
- ✓ Hypothesis Testing
- ✓ One Sample T test
- ✓ Normal Distribution
- ✓ Independent Sample T test
- ✓ P-value
- ✓ Chi-square test
- ✓ One-tailed and Two-tailed tests
- ✓ ANOVA

Capstone Project- 2

Machine Learning Specialization

Module-1: Introduction to Machine Learning

- ✓ What is Machine Learning?
- ✓ Applications of Machine Learning
- ✓ Machine Learning in your daily life
- ✓ Machine Learning in Retail
- ✓ Steps Involved in Machine Learning

Module-2: Regression

- ✓ Introduction to Regression
- ✓ Linear Regression
- ✓ Evaluation Metrics in Regression Models
- ✓ Logistic Regression
- ✓ **Mini Project**

Module-3: Supervised Classification

- ✓ Why Use Classification?
- ✓ Application of Classification Algorithms
- ✓ Introduction to Classification
- ✓ Types of Classification Algorithms
- ✓ Classification: Decision Tree
- ✓ Classification: Random Forest
- ✓ ML in Banking & Finance - Benefits
- ✓ Classification: SVM
- ✓ Classification: KNN
- ✓ Classification: Naïve Bayes
- ✓ Evaluating Classification Models
- ✓ Model Optimization Techniques
- ✓ Model Boosting Techniques
- ✓ Introduction to PyCaret
- ✓ Dealing with Unbalanced Datasets
- ✓ **4 Mini Projects**

Capstone Project- 3

Module-4: Unsupervised Learning

- ✓ What is Unsupervised Learning?
- ✓ Application of Unsupervised Learning
- ✓ Introduction to Clustering
- ✓ Types of Clustering
- ✓ Partitioning Methods: K-means, DBSCAN, Spectral
- ✓ Hierarchical Methods: Hierarchical
- ✓ **2 Mini Projects**

Module 5 :- Dimension Reduction

- ✓ PCA
- ✓ Factor Analysis
- ✓ LDA
- ✓ Mini Project

Module-6: Association Rules Mining

- ✓ What are Association Rules?
- ✓ Association Rule Parameters
- ✓ A-priori Algorithm
- ✓ Market Basket Analysis

Module-7: Recommendation System

- ✓ What is a Recommendation System?
- ✓ Need for a Recommendation System
- ✓ Recommendation System Use Cases
- ✓ Applications of Recommendation System
- ✓ Types of Recommendation Systems
- ✓ Collaborative Filtering
- ✓ Content Based Filtering
- ✓ Matrix Factorization
- ✓ Pros and Cons of Collaborative Filtering
- ✓ Hybrid Recommender System
- ✓ Mini Project

Module-8: Time-series Forecasting

- ✓ Introduction to forecasting data
- ✓ Properties of Time Series data
- ✓ Features of Time Series data
- ✓ Markov Processes - Overview and Terminologies
- ✓ Naive, Average and Moving Average Forecasting
- ✓ Exponential Smoothing
- ✓ ARIMA Approach
- ✓ Mini Project

Module 09 - Machine Learning Model Deployment

- ✓ Overview of Machine Learning Models
- ✓ Machine Learning System Architecture
- ✓ Research Environment
- ✓ Packaging, Serving and Deploying the model
- ✓ Differential Testing And More...

Capstone Project- 4

Deep Learning Specialization & AI

Module-1: Introduction to Deep Learning

- ✓ What is Deep Learning
- ✓ Curse of Dimensionality
- ✓ Machine Learning vs. Deep Learning
- ✓ Use Cases of Deep Learning
- ✓ Human Brain vs. Neural Network
- ✓ What is Perceptron?
- ✓ Learning Rate
- ✓ Epoch
- ✓ Batch Size

Module-2: Tensorflow 2.0 with Tensor Board

- ✓ Introduction to Tensorflow 2.x
- ✓ Installing Tensorflow 2.x
- ✓ Introduction to TensorBoard
- ✓ Defining Sequence model layers
- ✓ Activation Function
- ✓ Layer Types
- ✓ Model Compilation
- ✓ Model Optimizer
- ✓ Model Loss Function
- ✓ Model Training
- ✓ Digit Classification using Simple Neural Network in Tensorflow 2.x
- ✓ Mini Project

Module-3: Computer Vision

- ✓ Introduction to Convolutional Neural Networks
- ✓ Introduction to images
- ✓ Convolution, Pooling, Padding & its mechanisms
- ✓ Forward Propagation & Backpropagation for CNNs
- ✓ CNN architectures
- ✓ AlexNet, VGGNet, InceptionNet & ResNet
- ✓ Transfer Learning
- ✓ Object Detection

Module-4: Introduction to NLP

- ✓ Introduction to NLP
- ✓ Libraries & Tools for NLP in Python
- ✓ NLTK vs Spacy
- ✓ Applications of NLP:
- ✓ Chatbot
- ✓ Search, Autocorrect and Autocomplete
- ✓ Grammar Checker

Module-5: Text Processing Methods

- ✓ Bag of Words
- ✓ Countvectorizer
- ✓ Term Frequency (TF)
- ✓ Inverse Document Frequency (IDF)
- ✓ Converting text to features and labels
- ✓ Multinomial Naive Bayes Classifier
- ✓ Leveraging Confusion Matrix Assignment
- ✓ Word Embeddings
- ✓ Word2Vec
- ✓ **Mini Project**

Module-6: Introduction to Sequence Learning

- ✓ What is Sequence Learning
- ✓ Application of Sequence Learning
- ✓ What is Sequence Model
- ✓ Bayesian Network
- ✓ Markov Model
- ✓ Markov Chain
- ✓ Hidden Markov Model
- ✓ Viterbi Algorithm

Module-7: RNN vs LSTM

- ✓ Recurrent Neural Network
- ✓ Architecture of RNN
- ✓ Calculation in RNN
- ✓ Backpropagation and Loss calculation
- ✓ Applications of RNN
- ✓ What is LSTM?
- ✓ Structure of LSTM
- ✓ LSTM architecture
- ✓ **Mini Project**

Capstone Project- 5

Module-8: TensorFlow Hub for Object Detection using Faster RCNN

- ✓ Introduction to TensorFlow Hub
- ✓ Use cases of TensorFlow Hub
- ✓ Limitations of CNN in object detection
- ✓ Architecture of RCNN
- ✓ Applications of RCNN
- ✓ Types of RCNN
- ✓ Step by step implementation of Faster RCNN

Module-9: Sentiment Analysis

- ✓ Sentiment Analysis
- ✓ Subjectivity Analysis
- ✓ Topic Extraction
- ✓ Product Reviews
- ✓ Opinion Retrieval and Spam
- ✓ Opinion Summarization
- ✓ Implementing Sentiment Analysis in Python

Module-10: Reinforcement Learning

- ✓ Understanding Reinforcement Learning
- ✓ Algorithms associated with RL
- ✓ Q-Learning Model
- ✓ **Mini Project**

Module-11: Introduction to GANs (Generative adversarial networks)

- ✓ Introduction to GANs
- ✓ Generative Networks
- ✓ Adversarial Networks
- ✓ How do GANs work?
- ✓ DCGANs - Deep Convolution GANs
- ✓ Applications of GANs

Capstone Project- 6

Generative AI Specialization & ChatGPT, Prompt Engineering, Explainable AI

Module-1: Generative AI Specialization

- ✓ Introduction to Generative AI Models
- ✓ The Future of Generative AI
- ✓ Types of Generative AI Models
- ✓ Ethical Considerations in Generative AI Models & ChatGPT
- ✓ Popular Generative AI Models
- ✓ Benchmarking & Evaluating Models

Module-2: ChatGPT, Prompt Engineering, Explainable AI

- ✓ ChatGPT
- ✓ Fine-tuning ChatGPT
- ✓ Deploying and Scaling ChatGPT
- ✓ Maintaining ChatGPT
- ✓ Security and Privacy Considerations
- ✓ Monitoring and Debugging ChatGPT
- ✓ **Mini Project**

Explainable AI

Prompt Engineering

Advanced Prompt Engineering Techniques

ML & AI with IoT

Module 1: IoT Architecture, Protocols and ML

- ✓ IoT Architecture
- ✓ Signal Acquisition
- ✓ Data filtering and Pre-processing
- ✓ Inference handling
- ✓ Edge vs Cloud functional partitioning
- ✓ IoT Protocols
- ✓ HTTP
- ✓ MQTT
- ✓ CoAP
- ✓ 6LoWPAN

Module 2: ML Optimization Techniques

- ✓ Performance parameters
- ✓ Generic Optimization techniques
- ✓ Quantized NNs, and its Case Study:
8-bit Fixed Point Training
- ✓ Compression vis Early Exiting, Prediction,
and its Case Studies
- ✓ Tensorflow Lite

Hands-on:

- ✓ Tensorflow Lite
- ✓ 8-bit Fixed Point Training

Module 3: IoT Cloud Platforms supporting ML & AI

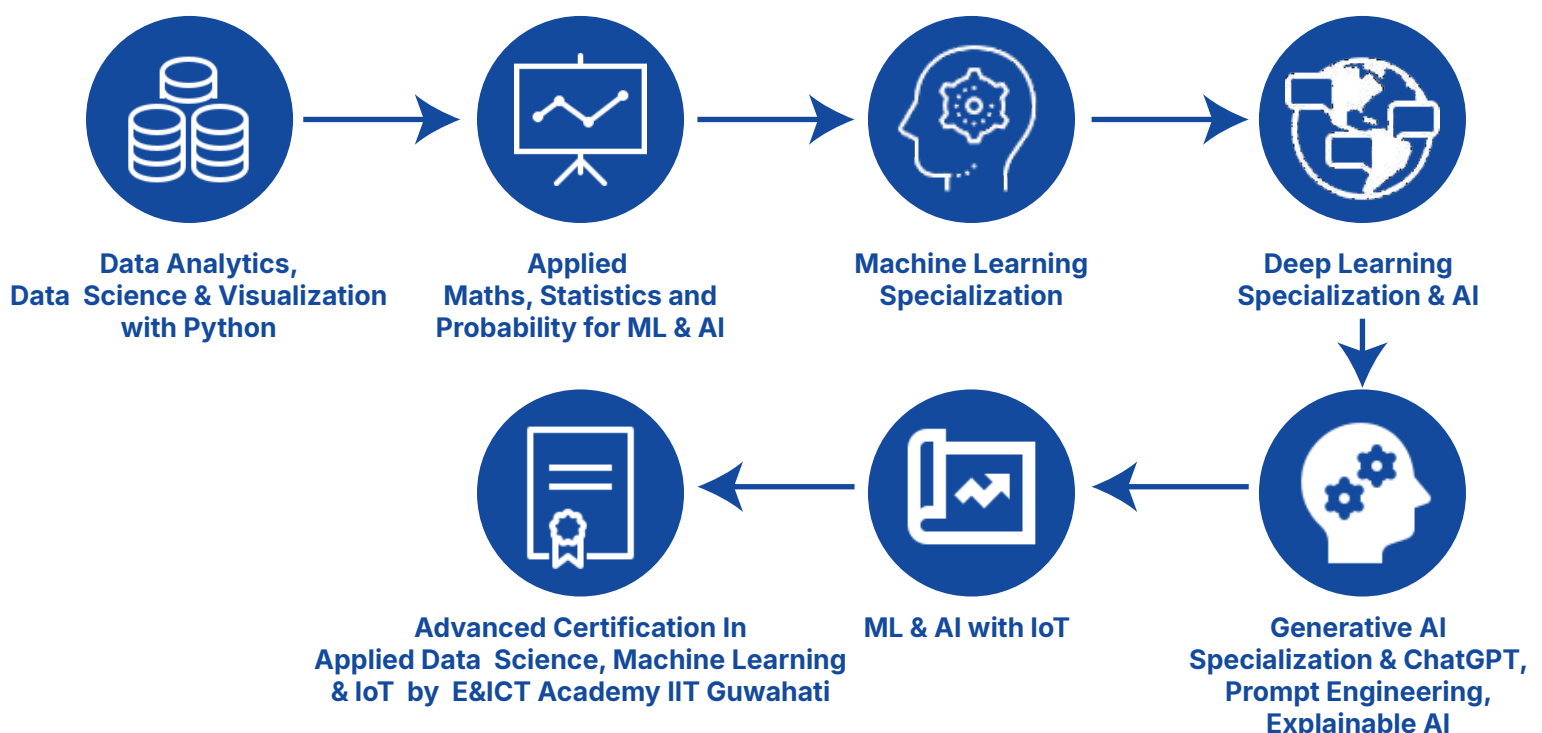
- ✓ Why Use Machine Learning (ML)
in the Cloud?
- ✓ ML Cloud Platforms
- ✓ Amazon ML vs Google Cloud
AutoML vs IBM Watson
- ✓ Overview of AWS SageMaker
- ✓ Machine Learning with AWS SageMaker
- ✓ Creating an account on AWS
- ✓ AWS- IOT Core
- ✓ Google Collab
- ✓ Build, Train, and Deploy a ML Model

Major Capstone Project

TOOLS, LANGUAGES & FRAMEWORKS

				IP[y]:

LEARNING PATH





Program Information:

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